Dual-Channel Prototype Network for Few-Shot Pathology Image Classification

Hao Quan, Xinjia Li, Dayu Hu, Tianhang Nan, and Xiaoyu Cui, *Member, IEEE*

***Abstract*—In the field of pathology, the scarcity of certain diseases and the difficulty of annotating images hinder the development of large, high-quality datasets, which in turn affects the advancement of deep learning-assisted diagnostics. Few-shot learning has demonstrated unique advantages in modeling tasks with limited data, yet explorations of this method in the field of pathology remain in the early stages. To address this issue, we present a dual-channel prototype network (DCPN), a novel few-shot learning approach for efficiently classifying pathology images with limited data. The DCPN leverages self-supervised learning to extend the pyramid vision transformer (PVT) to few-shot classification tasks and combines it with a convolutional neural network to construct a dual-channel network for extracting multi-scale, high-precision pathological features, thereby substantially enhancing the generalizability of prototype representations. Additionally, we design a soft voting classifier based on multi-scale features to further augment the discriminative power of the model in complex pathology image classification tasks. We constructed three few-shot classification tasks with varying degrees of domain shift using three publicly available pathological datasets—CRCTP, NCTCRC, and LC25000—to emulate real-world clinical scenarios. The results demonstrated that the DCPN outperformed the prototypical network across all metrics, achieving the highest accuracies in same-domain tasks—70.86% for 1-shot, 82.57% for 5-shot, and 85.2% for 10-shot setups—corresponding to improvements of 5.51%, 5.72%, and 6.81%, respectively, over the prototypical network. Notably, in the same-domain 10-shot setting, the accuracy of the DCPN (85.2%) surpassed that of the PVT-based supervised learning model (85.15%), confirming its potential to diagnose rare diseases within few-shot learning frameworks.**

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I. Introduction

HISTOPATHOLOGICAL slides provide crucial feature information for accurately diagnosing cancer [1]. Traditionally, pathologists identify tumor characteristics by directly observing tissue slides under a microscope, which is a process that is not only time-consuming but also susceptible to individual subjectivity[2]. Considering the complexity of classifying more than one hundred types of cancerous tissues, there is an urgent need for more efficient and precise computer-assisted diagnostic methods. The emergence of digital pathology has promoted significant developments in computational pathology [3], [4], where deep learning, particularly deep neural networks, has demonstrated remarkable potential in medical image analysis scenarios. However, the availability of large-scale labeled data is a key prerequisite for the superior performance of deep learning methods [5]. This stands in contrast to real-world clinical settings, where high-quality, labeled medical imaging data are challenging to obtain [6].

Few-shot learning (FSL) is dedicated to addressing the challenge of data scarcity and has made significant progress in the field of natural image analysis [7], [8], [9]. In few-shot classification tasks, the idea is that classifiers usually learn from a limited number of samples and handle new classes not seen during the training process. Data augmentation-based FSL methods, primarily implemented through generative adversarial networks (GANs) [10] and mixup data augmentation [11], are used to acquire additional training samples to address the issue of insufficient data. However, the data generated by these methods may exhibit distributional biases and may not accurately reflect the true data distribution, potentially reducing the generalizability of the constructed model. Meta-learning approaches, such as model-agnostic meta-learning (MAML) [8], focus on the model initialization procedure to facilitate rapid adaptation to new tasks after a few gradient updates. However, the training process involves complex second-order gradient calculations and storage of multiple gradient states, leading to high computational and memory costs. Metric learning approaches, such as siamese networks [12], matching networks [13], and prototypical networks [14], attain enhanced generalizability by optimizing distance metrics between samples and have computational efficiency and training stability advantages over data augmentation and

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meta-learning. These methods are particularly effective in the field of medical image analysis because they can capture subtle visual differences and adapt to common class imbalance and sample scarcity issues.

In the field of medical image analysis, FSL methods are widely applied to classification and segmentation tasks involving CT and MRI images [15], [16], [17], [18], but less research has been conducted on pathology images. Studies concerning pathology images mainly focus on metric learning-based Siamese networks and prototypical networks. Siamese networks achieve classification by measuring the similarity between two inputs through multiple subnetworks with the same structure and shared parameters. Medela et al. [19] constructed a Siamese network with three backbones for the classification of pathological colon, lung, and breast tissue images, and demonstrated that Siamese networks are significantly superior to model fine-tuning when the input data are insufficient. Conversely, prototypical networks [14] use an embedding network to cluster the feature embeddings of samples located near the prototype representations of their respective categories. Shaikh et al. [20] used prototypical networks to filter pathology images containing artifacts, optimizing the analysis of downstream tasks. Deuschel et al. [21] expanded the single prototype of Prototypical Networks to multiple prototypes, an improvement that extended the feature space of prototype representations, thereby enhancing the rapid adaptability of networks to new data. Expanding the feature space of prototype representations is considered key to improving the generalizability of prototypical networks and achieving superior performance.

Currently, convolutional neural networks (CNNs) are widely used as embedding networks for prototypical networks. CNNs struggle to encode the long-range dependencies within images and lack global modeling capabilities, which limits the generalizability of their prototype representations [22], [23], [24]. In contrast, the vision transformer (ViT) [25] has demonstrated significant global feature modeling advantages for visual recognition tasks and is a promising method for further improving the generalizability of prototype representations. Gan et al. [26] constructed global features by encoding the correlations between the context features within the support set through transformer blocks. However, this approach only uses transformer blocks to integrate the local features extracted by CNNs and does not directly extract global features from the input images. Indeed, due to the data-hungry nature of ViTs, it is challenging to directly apply them to FSL methods that aim to solve the sample insufficiency problem [27]. How to extend the ViT architecture to few-shot classification tasks based on prototypical networks and improve the resulting prototype representations remains a question worth exploring.

Based on the aforementioned analysis, this study constructs a dual-channel prototype network (DCPN) that integrates a transformer and CNNs, successfully extending the transformer model to few-shot classification tasks. First, we employ a self-supervised learning strategy to pretrain the transformer model, enabling it to learn discriminative feature representations on a large-scale unlabeled pathology image dataset to enhance its generalizability and reduce the likelihood of overfitting. Next, by combining it with CNNs, which have inherent inductive biases, the resulting model can effectively capture the local features of images. Subsequently, we merge the features encoded by the transformer and CNNs to output multi-scale features, thereby improving the generalizability of the prototype representations. Finally, we employ a soft voting strategy to construct a robust classifier [28]. Furthermore, we introduce a regularization strategy to mitigate the overfitting problem encountered by transformers in FSL [29]. In summary, the main contributions of this paper are as follows

1) We propose an innovative method for conducting few-shot classification tasks in pathology images, the dual-channel prototype network (DCPN), which effectively addresses the modeling challenges caused by sample scarcity and has potential applications concerning the auxiliary diagnosis of rare diseases.
2) We construct a multi-scale prototype representation space based on transformers and CNNs, significantly enhancing the generalizability of the prototype features.
3) We design a soft voting classifier based on multi-scale features, enhancing the robustness of the model classification process.
4) Through the design of few-shot pathology image classification tasks with three different degrees of domain shift, we demonstrate the effectiveness of the proposed method and verify its applicability in other label-hungry problems.

II. Related Work

FSL aims to adapt models to new tasks or categories using the smallest possible set of examples [7]. This characteristic reveals its substantial potential for addressing data-scarce scenarios. This is particularly true within the context of medical image analysis, where it offers new possibilities for handling diverse and limited patient data, especially for modeling rare diseases. In this paper, we introduce the dual-channel prototype network (DCPN) method, which integrates ideas derived from transfer learning, metric learning, and meta-learning approaches into few-shot learning. To this end, we conduct a comprehensive and in-depth survey of the literature from these three perspectives.

A. Methods Based on Transfer Learning

Transfer learning-based FSL strategies primarily focus on acquiring knowledge from source tasks and transferring it to target tasks with scarce samples [30]. Within the realm of medical image analysis, these FSL methods have achieved significant advancements [31], [32], [33], [34], [35]. For instance, Spolaôr et al. [31] demonstrated the efficacy of implementing transfer learning to model small datasets by fine-tuning CNN models pretrained on the ImageNet dataset, successfully adapting them to classify dermatoscopic images and COVID-19 CT images. However, due to the intrinsic differences between natural and medical images, this approach has limitations when addressing the challenge of conducting precise auxiliary diagnoses for rare diseases. To overcome these constraints, Chen et al. [32] confirmed the effectiveness of transfer learning-based FSL

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methods in diagnosing rare diseases by pretraining networks with a substantial amount of base class data. This method capitalizes on the similarities between diseases, concatenating the features of base and novel classes, thereby enabling the model to generalize from well-sampled base classes to specific novel class tasks. Nevertheless, when the source and target tasks exhibit a low degree of similarity, domain bias can lead to negative transfer, and overfitting issues may also impair the resulting model performance [32].

B. Methods Based on Metric Learning

Metric learning-based FSL methods are designed to learn a distance function that can measure the similarity between samples [13]. This approach maps images into a space where the distance metric is associated with the similarity between categories, enabling classification by comparing new samples with a set of labeled samples [36]. Specifically, matching networks, introduced by Vinyals et al. [13], employ a novel architecture that combines attention mechanisms and memory components to directly compute the distances between a test sample and all samples in the support set, thus enabling effective instance-based learning. Then, the prototypical networks proposed by Snell et al. [14] further streamlined the process by representing each category in the embedding space with the mean vector of its examples and performing classification based on the nearest prototype. The applications of metric learning, including the direct application of metric models [15], [37], [38], [39], the construction of multi-learner systems in conjunction with meta-learning [16], [40], and the use of methods combined with self-supervised learning, are widespread in the field of medical image analysis [41]. The key to the success of this type of method lies in utilizing a precise distance measurement, and capturing the critical feature differences to enhance its generalizability remains the primary challenge.

C. Methods Based on Meta-Learning

Meta-learning, also known as learning to learn, is centered on the concept of training models to learn from a limited set of samples across multiple tasks, enabling rapid adaptation to new tasks [36]. A quintessential method in this domain is model-agnostic meta-learning (MAML), which was proposed by Finn et al. [8]. This method seeks to identify an optimal set of initial parameters through training across multiple tasks, and the constructed model can be swiftly adapted to new tasks with a few gradient descent steps. Subsequently, Lee et al. [42] enhanced MAML by adopting differentiable convex optimization to improve the generalizability of the model with limited data. In the domain of medical image analysis, the MAML approach has achieved notable progress [43], [44], [45]. However, its requirement of computing second-order derivatives during the iterative process leads to substantial computational resource consumption. Consequently, researchers have explored meta-learning methods based on transfer learning [46], [47] and metric learning [16], [40], which have increased computational efficiency and training stability. In summary, the core challenge of meta-learning lies in achieving rapid model adaptability and generalization across tasks.

By integrating the aforementioned methods, this article introduces an innovative dual-channel prototype network (DCPN) for few-shot classification tasks involving pathology images. Our approach employs self-supervised learning to encode rich pathological features and initialize the model parameters and harnesses the principles of meta-learning to train the dual-channel model for rapidly adapting it to new tasks. Furthermore, we design a soft voting strategy based on multi-scale features to construct a metric function, thereby enhancing the discriminative power of the model in few-shot scenarios.

III. Problem Definition

The long-tailed issue of disease data has resulted in the majority of illnesses having only a limited amount of data, leading to a lack of large-scale annotated datasets. This issue constrains the applicability of traditional supervised deep learning methods in medical imaging. Additionally, the emergent nature of diseases demands that deep learning models possess flexible scalability. Consequently, we define the pathology image classification task using the FSL paradigm. Generally, FSL refers to the algorithmic approach where a model is trained on a base dataset $D_{base} = \{(x_i, y_i)\}_{i=1}^{N_{base}}$ to develop discriminative capabilities, which are then generalized to recognize new classes with only a few labeled samples $D_{novel} = \{(x_t, y_t)\}_{t=1}^{N_{novel}}$, where the label spaces $y_i \cap y_t = \emptyset$. We attempt to conceptualize FSL as an N-way, K-shot meta-learning framework, employing an episodic approach to accomplish FSL for pathology images. Meta-learning is divided into meta-training and meta-testing phases, each requiring the construction of numerous meta-tasks; Fig. 1 providing an example of a 5-way, 1-shot meta-learning scenario. The meta-tasks consist of a support set S and a query set Q and are denoted as $T = \{(S_j, Q_j)\}_{j=1}^{N_{task}}$, where N_{task} is the number of meta-tasks sampled per epoch. The data for constructing the meta-tasks in the meta-training and meta-testing phases are sampled from D_{base} and D_{novel}, respectively. If the meta-tasks for both meta-training and meta-testing are derived from $D_{joint} = D_{base} \cap D_{novel}$, we refer to this type of FSL as generalized few-shot learning (GFSL).

IV. Methodology

We propose a multi-scale prototype network approach for FSL classification tasks involving pathology images. Fig. 2 illustrates the overall workflow of this method, which primarily encompasses three parts: 1. self-supervised pretraining of the PVT network; 2. The construction of a dual-channel prototype network to extract multi-scale prototype representations; and 3. The development of a robust classifier based on a soft voting strategy.

A. Self-Supervised Pretraining Process of the PVT Model

The global feature modeling capabilities of vision transformers have been demonstrated to provide significant advantages in

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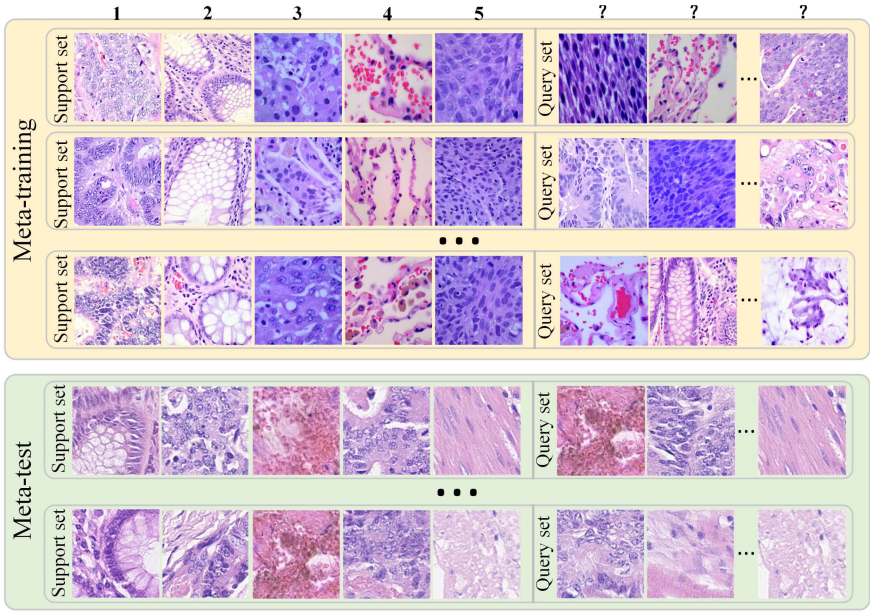

Fig. 1. Examples of meta-tasks within the meta-learning framework. Each row represents a meta-task composed of a support set and a query set, corresponding to a 5-way, 1-shot few-shot pathology image classification task.

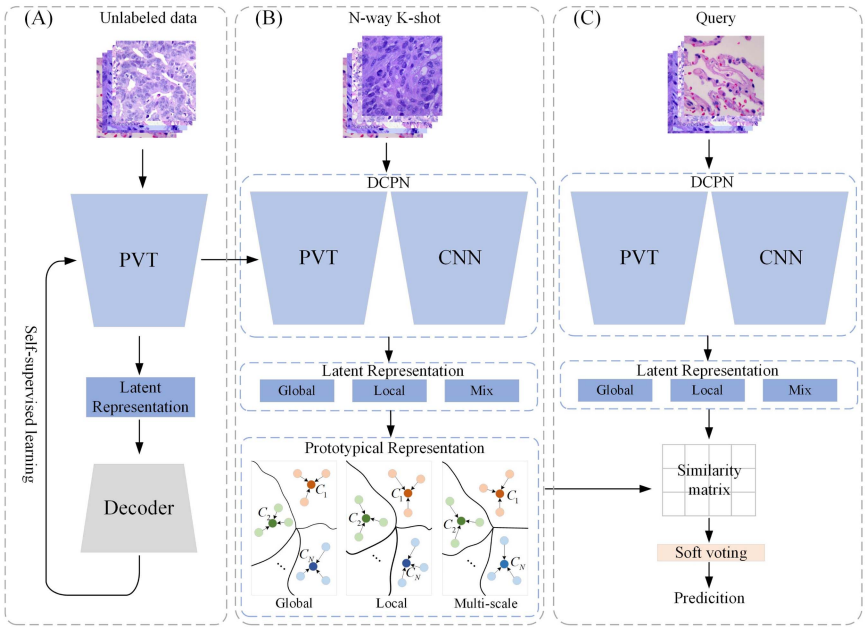

Fig. 2. Overview of the few-shot pathology image classification process constructed using the DCPN method. Initially, (A) the PVT model is pretrained based on self-supervised learning. Subsequently, (B) a dual-channel network is constructed in conjunction with a CNN to extract multi-scale features. Finally, (C) few-shot classification is realized using a similarity matrix based on multi-scale features and coupled with a soft voting strategy.

multiple subdomains of computer vision. Among these methods, the PVT [48], an improved version of the original ViT, aims to reduce the computational complexity of multi-head self-attention by introducing a pyramid structure, thereby capturing richer contextual features. Hence, in this paper, the PVT network is chosen as the encoder for representing the global features of pathology images.

The masked autoencoder (MAE) [49] exhibits outstanding performance across multiple visual tasks through its asymmetric encoder-decoder architecture. This achievement is partly attributed to the ability of the vanilla vision transformer to extract global features. However, the PVT network typically introduces operations within local windows to reduce the quadratic complexity of the global self-attention process, which can

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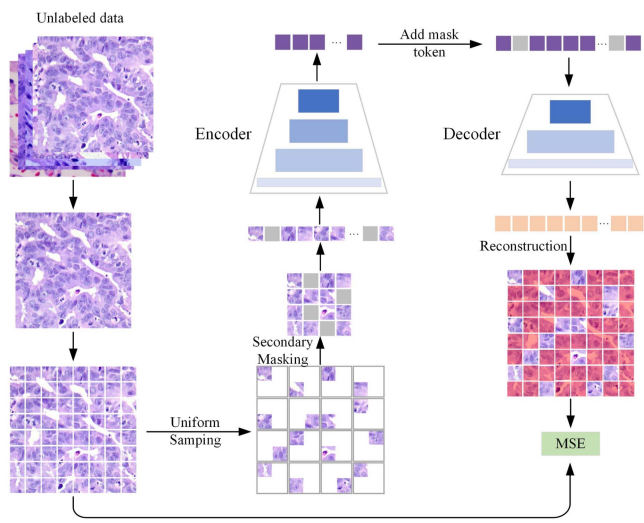

Fig. 3. Architectural diagram of the MAE based on the Uniform masking strategy.

completely lose the visual token information contained within these randomly constituted visual token sequences. To address this issue, our study introduces a uniform masking strategy [50], which improves upon the random masking method in the MAE, thus enabling the effective application of the MAE architecture to the pretraining process of the PVT.

Specifically, uniform masking includes two masking stages: uniform sampling (US) and secondary masking (SM). As illustrated in Fig. 3, the original input image is first partitioned into chunks; then, during the US phase, 25% of the patches are sampled from the image based on a uniform distribution strategy, ensuring that at least one patch is selected in every 2×2 grid across the entire image, resulting in a uniformly distributed set of sparse image blocks. SM further randomly masks a portion (approximately 25%) of the sampled areas without completely discarding the masked blocks. Instead, these blocks are treated as a shared, learnable token sequence and combined with the visible visual tokens as the encoder input.

The designs of the encoder, decoder, and reconstruction target follow the original setup of the MAE. However, unlike the original MAE, the encoder here is a PVT containing four stages, each of which outputs feature maps with different resolutions and a total stride of 32. This pyramid-structured transformer encodes the input 16×16 patches into subpixel-level latent representations. Therefore, before inputting the patches into the decoder, a PixelShuffle operation is required to restore the resolution of the latent features. Finally, the decoder reconstructs the missing pixels of the original image and uses the mean squared error (MSE) as the loss function to optimize the model; the formula of the MSE is as follows:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}\left(Y_i - \hat{Y}_i\right)^2 \tag{1}$$

where N is the total number of pixels missing from the original image, Y_i is the true pixel value of the i-th pixel, and $\hat{Y}_i$ is the pixel value predicted by the model. By minimizing the MSE loss, the model is trained to accurately reconstruct each pixel, thereby learning the global features of the input image.

B. Construction of Multi-scale Prototype Representations

Multi-scale prototype representations are acquired through a dual-channel encoder, as shown in Fig. 4(a). We utilize a pretrained PVT to extract the global features of pathology images, while a CNN is employed to extract local features. Specifically, for a given input image x:

$$Z_G = f_{PVT}(x) \tag{2}$$

$$Z_G = f_{CNN}(x) \tag{3}$$

Here, Z_G and Z_L represent the global and local feature embeddings of the input image x, respectively, each possessing D dimensions. To further refine these features and reduce their dimensionality, we apply principal component analysis (PCA) [51] to process the global and local feature embeddings, extracting a set of linearly independent feature embeddings and removing as much redundant information as possible. The processed features are then concatenated, resulting in mixed semantic features:

$$Z'_G, Z'_L = PCA(Z_G, Z_L) \tag{4}$$

$$Z_{Mix} = Concat(Z'_G, Z'_L) \tag{5}$$

Here, Z'_G and Z'_L are the global and local features obtained after the PCA dimensionality reduction process, respectively, each with $D/2$ dimensions. Z_{Mix} denotes the mixed semantic features obtained by concatenating Z'_G and Z'_L, possessing a dimensionality of D. At this point, the input image x can be represented as a collection of three different scale feature sets $\{Z_G, Z_L, Z_{Mix}\}$.

Next, to construct the multi-scale prototype representation, we consider a support set $D_{support}$ containing N categories. For each category C_i(where $i \in \{1, 2, \ldots, N\}$), the input is a set of images $\{x_1, x_2, \ldots, x_K\}$. Here, K denotes the number of samples per category in the support set, i.e., the K-shot setting. Each image is mapped to the multi-scale feature space, resulting in $\{Z^j_G, Z^j_L, Z^j_{Mix}\}_{j=1,2,\ldots,K}$. For each category C_N, we compute its multi-scale prototype representation in the corresponding feature space, which is the mean of all image features in the support set:

$$MP_{C_i} = \left\{\frac{1}{K}\sum_{j=1}^{K} Z^j_G, \frac{1}{K}\sum_{j=1}^{K} Z^j_L, \frac{1}{K}\sum_{j=1}^{K} Z^j_{Mix}\right\} \tag{6}$$

Here, MP denotes the multi-scale prototype. Thus, each category C_i is represented by a set of multi-scale prototype representations $MP_{C_i} = \{P^{C_i}_G, P^{C_i}_L, P^{C_i}_{Mix}\}$. Ultimately, we obtain

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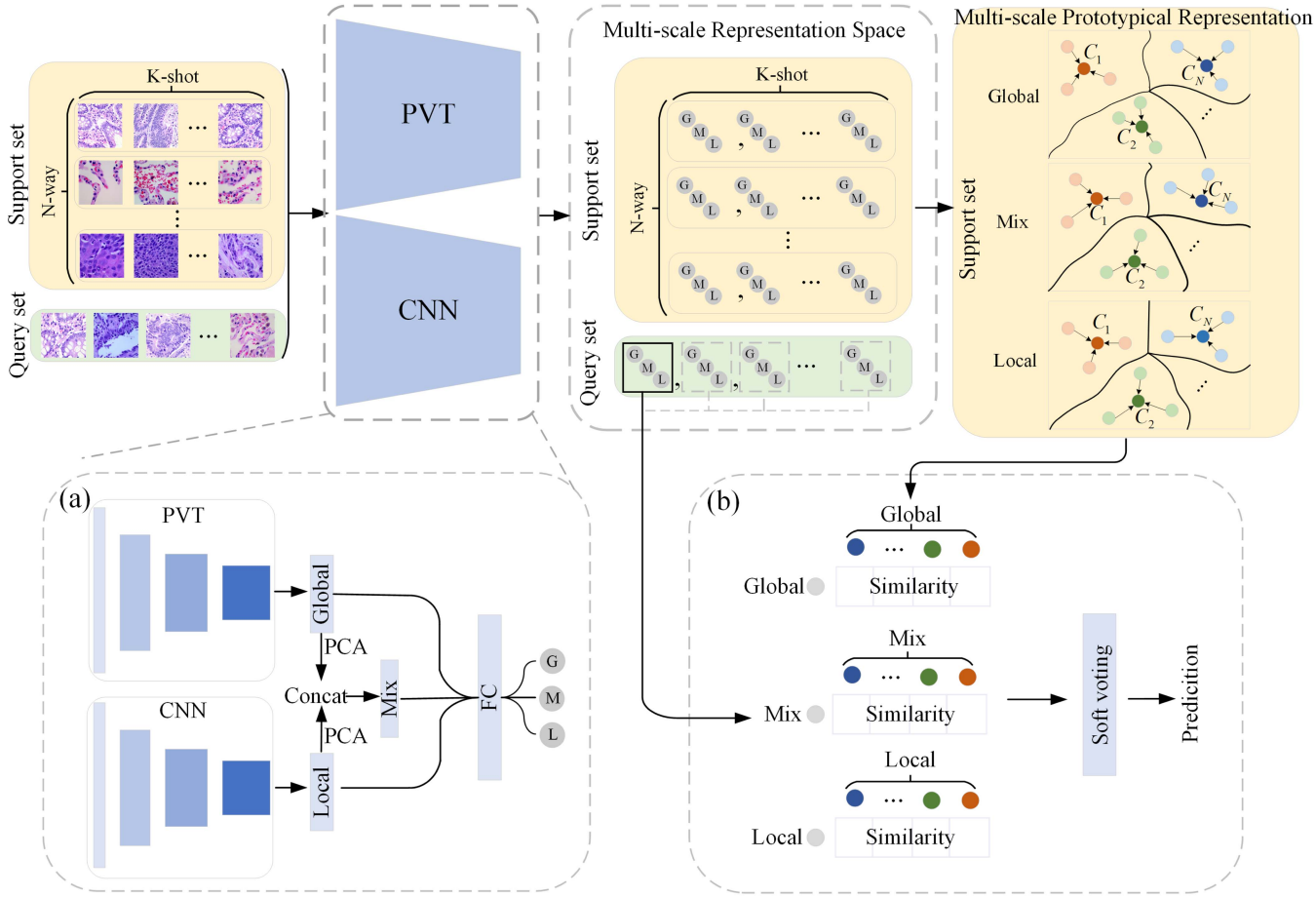

Fig. 4. Dual-channel prototype network. (a) The dual-channel network for multi-scale feature construction; (b) the soft voting classifier built upon multi-scale features.

a multi-scale prototype representation matrix:

$$MP = \begin{bmatrix} P_G^{C_1} & P_L^{C_1} & P_{Mix}^{C_1} \\ P_G^{C_2} & P_L^{C_2} & P_{Mix}^{C_2} \\ \vdots & \vdots & \vdots \\ P_G^{C_N} & P_L^{C_N} & P_{Mix}^{C_N} \end{bmatrix}_{N\times 3} \tag{7}$$

This matrix reflects the average features of each category sample in the support set across different scales, aiming to provide a rich and effective prototype representation for the pathology image classification task in FSL scenarios.

C. Multi-scale Feature Soft Voting Classifier

The soft voting strategy, which is a common classification method in ensemble learning [28], [52], [53], is typically based on the probability prediction results produced by multiple classifiers for casting votes. It takes the predicted probabilities of each base classifier for the different classes into full account, which can enhance the performance of the classifier. Unlike traditional soft voting methods, in this paper, we employ a soft voting strategy designed with multi-scale features using a single classifier, aiming to avoid the overfitting issues that may arise when training multiple classifiers, as shown in Fig. 4(b).

Given a query set sample x_q, we first extract a set of multi-scale feature sets $\{Z_G^q, Z_L^q, Z_{Mix}^q\}$ through a dual-channel encoder. To determine the category of x_q, we calculate the Euclidean distance between the query sample and the prototypes of each category in the multi-scale prototype representation matrix at the corresponding scale, as follows:

$$\begin{bmatrix} d_G^{C_i} \\ d_L^{C_i} \\ d_{Mix}^{C_i} \end{bmatrix} = \begin{bmatrix} \left\| Z_G^q - P_G^{C_i} \right\|_2 \\ \left\| Z_L^q - P_L^{C_i} \right\|_2 \\ \left\| Z_{Mix}^q - P_{Mix}^{C_i} \right\|_2 \end{bmatrix} i \in \{1, 2, \ldots, N\} \tag{8}$$

Where $d_G^{C_i}$, $d_L^{C_i}$, and $d_{Mix}^{C_i}$ represent the Euclidean distances between the global, local, and mixed features of the query sample x_q and the prototype representation of the C_ith category at the corresponding scale, respectively.

To convert the distances into confidence values, we use a negative exponential function to prevent the numerical instability caused by distances approaching zero. The confidence level for each scale is calculated through the negative exponential of the

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associated distance:

$$\alpha_{C_i} = \exp\left(-d_G^{C_i}\right) + \exp\left(-d_L^{C_i}\right) + \exp\left(-d_{Mix}^{C_i}\right) \quad (9)$$

Subsequently, we normalize these confidence values using the softmax function to calculate the probability of the query sample x_q belonging to each category:

$$p(C_i|x_q) = \frac{\exp(\alpha_{C_i})}{\sum_{j=1}^{N}\exp(\alpha_{C_j})} \quad (10)$$

Finally, we select the category with the highest probability as the prediction result:

$$y = \arg\max_{C_i} p(C_i|x_q) \quad (11)$$

where y represents the predicted category. The optimization process for model training employs the following loss function:

$$Loss(x_q, y_q) = -\log(p(C_i|x_q)) \quad (12)$$

where y_q is the true label corresponding to x_q. Algorithm 1 provides the pseudocode for the DCPN.

Algorithm 1: Dual-Channel Prototype Network (DCPN). N is the Number of Classes in Each Episode, K is the Number of Samples in Each Class, and Random_Sample(D, K) Denotes Randomly Selecting K Samples From the Set D.

Input: Support set $D_{support}$ and query set D_{query}, both comprising N classes; feature extractors f_{PVT} and f_{CNN}, where f_{PVT} is initialized with parameters from self-supervised pre-training process.
Output: Class prediction for the query samples

While not converged **do**
  For each episode **do**
    For $c \in N$ **do**
      $S_c \leftarrow$ Random_Sample($D_{support}$, $K_{support}$)
      $Z_G^c, Z_L^c \leftarrow \frac{1}{K}\sum_{x_i \in S_c} f_{PVT}(x_i), \frac{1}{K}\sum_{x_i \in S_c} f_{CNN}(x_i)$
      $Z_{Mix}^c \leftarrow Concat(PCA(Z_G^c), PCA(Z_L^c))$
      $MP_c \leftarrow \{Z_G^c, Z_L^c, Z_{Mix}^c\}$
    End for
    $Q \leftarrow$ Random_Sample(D_{query}, K_{query})
    $j \leftarrow 0$ // Initialize the loss
    For all $(x_q, y_q) \in Q$ **do**
      $Z_G^q, Z_L^q \leftarrow f_{PVT}(x_q), f_{CNN}(x_q)$
      $Z_{Mix}^q \leftarrow Concat(PCA(Z_G^q), PCA(Z_L^q))$
      For $c \in N$ **do**
        $d_G^c, d_L^c, d_{Mix}^c \leftarrow \left\|Z_G^q - Z_G^c\right\|_2, \left\|Z_L^q - Z_L^c\right\|_2, \left\|Z_{Mix}^q - Z_{Mix}^c\right\|_2$
        $\alpha_c \leftarrow \exp(-d_G^c) + \exp(-d_L^c) + \exp(-d_{Mix}^c)$
        $p(c \mid x_q) = \frac{\exp(\alpha_c)}{\sum_{j=1}^{N}\exp(\alpha_{c_j})}$ // Calculate the probability of the sample x_q being predicted as belonging to class c
        $j \leftarrow j + [-\log(p(c \mid x_q))]$ // Update loss
      End for
    End for
    $j \leftarrow j / K_{query}$ and update parameters θ of the DCPN model using Adam optimizer
  End for

V. Experiments

A. Datasets

This study selected three public histology datasets for model evaluation purposes. During the dataset division process, we utilized a stratified sampling strategy to ensure that the compositions of the training and test sets matched the distribution of the various tissue types found in the original datasets. For the NCTCRC dataset, we adopted the official partitioning settings provided. The details of the datasets are as follows:

- *CRCTP dataset [54]:* This dataset includes 280000 nonoverlapping patches with sizes of 150 × 150, which were extracted at 20 × magnification from 20 H&E-stained colorectal cancer whole-slide images (WSIs), encompassing seven different tissue types: tumors, stroma, complex stroma, muscle, debris, inflammatory, and benign. Within this dataset, 70% of the data were used as the training set, and 30% were used as the test set.
- *NCTCRC dataset [55]:* Comprising 107180 nonoverlapping patches with sizes of 224 × 224 extracted from 86 H&E-stained colorectal cancer WSIs, this dataset includes nine different tissue types: adipose (ADI), background (BACK), debris (DEB), lymphocytes (LYM), mucus (MUC), smooth muscle (MUS), normal colon mucosa (NORM), cancer-associated stroma (STR), and colorectal adenocarcinoma epithelium (TUM). Here, 100000 patches were designated for training, with the remaining 7180 for testing. Although both the NCTCRC and CRCTP originate from colorectal sources, they differ in their disease categories and regional data sources.
- *LC25000 dataset [56]:* This dataset contains 25000 patches with sizes of 768 × 768, including five different tissue types: colon adenocarcinomas, benign colonic tissues, lung adenocarcinomas, lung squamous cell carcinomas, and benign lung tissues, with 5000 patches for each tissue type. In this dataset, 70% of the data were allocated for training, and 30% were allocated for testing.

The diversity of the disease and sources among these three datasets allowed us to construct tasks with varying degrees of domain shifts. Specifically, we defined three different data realms based on the origins of the new classes and base classes. When the new and base classes came from the same organ and data source, they were considered to belong to the same domain. When the new and base classes originated from the same

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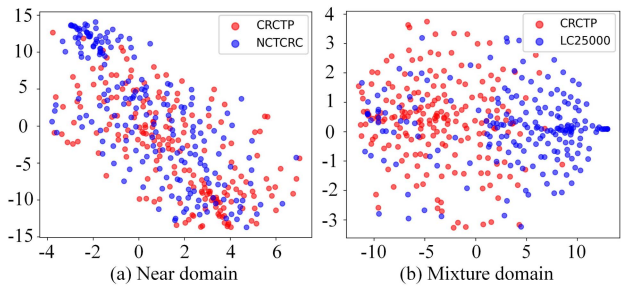

Fig. 5. Data Distribution for the near-domain and mixed-domain tasks.

organ but from different data sources, they were categorized as belonging to nearby domains. When the new and base classes were not derived from the same organ or from the same data source, they were defined as belonging to a mixed domain. The specific task settings were as follows:

- *Same-domain task:* The new and base classes both came from the CRCTP dataset, defining a 7-way, n-shot task. In this task, meta-training and meta-testing were carried out on the training and testing sets of the CRCTP dataset, respectively.
- *Near-domain task:* The CRCTP dataset was utilized as the base dataset for meta-training, and the NCTCRC dataset was utilized as the new class dataset for meta-testing; specifically, this formed a 5-way, n-shot task.
- *Mixed-domain task:* Similar to the near-domain task, CRCTP was used as the base dataset for meta-training, but LC25000 was chosen as the new class dataset for meta-testing; specifically, this formed a 5-way, n-shot task. Notably, within LC25000, two classes are related to colorectal tissues, and three are related to pulmonary tissues; therefore, LC25000 was defined as a mixed domain.

We employed the t-SNE method to reduce the datasets for the near-domain and mixed-domain tasks to three-dimensional space for visualization, observing the data distributions under different tasks, as shown in Fig. 5. To quantitatively assess the difficulty of the tasks, we calculated the Euclidean distance between the datasets. Specifically, the Euclidean distance between the CRCTP and NCTCRC datasets in the near-domain task was 0.20, while in the mixed-domain task, the distance between the CRCTP and LC25000 datasets was 1.01. This indicates that the difficulty of the mixed-domain task was significantly greater than that of the near-domain task, remaining consistent with our task design expectations.

B. Experimental Setup

During the pre-training phase, our study utilized the UM-MAE algorithm to pretrain a PVT as the global feature encoder for the training sets of the aforementioned three datasets, aiming to mitigate the difficulty of training during FSL tasks. The size of the input images was adjusted to 224×224, with a batch size set to 256 and an efficient batch size of 1024. Optimization was carried out using the AdamW optimizer with betas $= (0.9, 0.95)$, a learning rate of 1×10^{-3}, and 100 epochs. The remaining parameters were kept consistent with those of the original UM-MAE algorithm.

In the FSL phase, the sizes of the input images remained at 224×224, with a batch size of $N \times K$ (depending on the specific N-way, K-shot task), and optimization was performed using the Adam optimizer with a learning rate of 1×10^{-4} and 100 epochs. For the evaluation phase, 1000 meta-tasks were randomly drawn for each task to assess the model, with each meta-task utilizing 15 samples as a query set and accuracy employed as the evaluation criterion. All results were the mean outcomes of 1000 meta-tasks. To address the imbalance between the numbers of base classes and new classes, we followed the settings detailed in [57].

C. Comparison With Other SOTA Methods

In this experiment, we compared various advanced few-shot learning methods to validate the performance of the proposed approach. These algorithms included four classic few-shot learning algorithms, namely, the matching network [13], ProtoNet [14], MetaOpt [42], and MAML [8]; three pathology-specific few-shot learning algorithms, namely, Latent Augmentation [58], the histology Siamese network [19], and Multi-ProtoNets [21]; and two state-of-the-art few-shot learning algorithms, namely, CGRN [59] and LDP-Net [60]. Notably, LDP-Net and the histology Siamese network are typical multi-channel few-shot algorithms. The experimental results are shown in Table I.

The experimental results indicate that the DCPN significantly outperformed the other classic few-shot learning methods in terms of accuracy. In the same-domain, 10-shot scenario, the DCPN achieved an accuracy of 85.2%, even surpassing that of the PVT-based supervised learning model (85.15%) (as shown in Table II). Compared to the 74.69% of the matching network, the 78.39% of ProtoNet, the 74.21% of MetaOpt, and the 71.32% of MAML, the proposed approach exhibited improvements of 10.51%, 6.81%, 10.99%, and 13.88%, respectively. Among the three pathology-specific methods, the DCPN still maintained the highest accuracy. The latent augmentation method, which introduces latent variables to increase the diversity of data to adapt to new categories, may introduce noise through random variables, limiting its performance in few-shot learning tasks. The histology Siamese network applies a Siamese network directly to few-shot classification tasks involving pathology images, but its effectiveness was limited due to a lack of in-depth adjustments for the characteristics of pathology images. Multi-ProtoNets also achieved competitive results by constructing multiple prototype representations through k-means clustering. CGRN and LDP-Net, the latest few-shot learning algorithms, have also demonstrated good adaptability in the field of pathology. Among the mentioned methods, LDP-Net and the histology Siamese network, through parameter sharing or momentum updates, constructed a pseudo multichannel architecture, which still had limited generalizability in the feature space compared to that of the DCPN. Additionally, the computational complexity of the DCPN was only 10.02 GFLOPs, which was significantly lower than the 33.03 GFLOPs of the classic ViT-B/16 model, making

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TABLE I
COMPARISON AMONG THE STATE-OF-THE-ART METHODS

Method	Backbone	Same domain			Near domain			Mixed domain		
		1-shot	5-shot	10-shot	1-shot	5-shot	10-shot	1-shot	5-shot	10-shot
Matching network[13]	ResNet50	62.47 ±1.36	72.85 ±1.58	74.69 ±0.97	60.98 ±1.91	71.26 ±1.42	72.08 ±1.07	50.41 ±1.21	55.74 ±1.32	57.82 ±1.58
ProtoNet[14]	ResNet50	65.35 ±0.89	76.85 ±0.95	78.39 ±0.84	61.81 ±0.81	72.75 ±0.72	73.98 ±0.85	**53.91 ±0.68**	**59.45 ±0.91**	60.96 ±0.75
MetaOpt[42]	ResNet50	63.32 ±0.59	70.08 ±0.77	74.21 ±0.92	59.20 ±0.84	67.83 ±0.75	70.74 ±0.69	52.82 ±0.68	54.81 ±0.75	56.96 ±0.88
MAML[8]	ResNet50	60.32 ±1.21	68.41 ±1.42	71.32 ±0.98	58.23 ±1.22	66.45 ±1.47	70.89 ±1.32	52.18 ±1.58	54.23 ±1.69	58.97 ±1.77
Latent Augmentation[58]	ResNet50	66.92 ±0.69	79.63 ±0.64	82.41 ±0.93	62.10 ±0.81	72.09 ±0.47	75.61 ±0.52	53.31 ±0.68	57.19 ±0.62	63.73 ±0.84
Histology Siamese Network[19]	ResNet50	63.21 ±0.49	75.82 ±0.34	79.43 ±0.52	61.85 ±0.71	71.46 ±0.66	74.31 ±0.64	52.02 ±0.58	55.42 ±0.47	61.36 ±0.39
Multi-ProtoNets[21]	ResNet50	67.91 ±0.55	**80.45 ±0.47**	**85.12 ±0.58**	**62.98 ±0.62**	73.42 ±0.57	**76.67 ±0.61**	52.87 ±0.59	58.45 ±0.61	**65.21 ±0.47**
CGRN[59]	ResNet50	63.87 ±0.93	74.58 ±0.56	79.24 ±0.76	61.82 ±0.61	70.12 ±0.47	72.85 ±0.81	51.23 ±0.74	57.85 ±0.58	59.41 ±0.41
LDP-Net[60]	ResNet50	**68.43 ±0.52**	79.91 ±0.37	83.16 ±0.67	62.1 ±0.33	**74.27 ±0.61**	76.4 ±0.43	53.7 ±0.33	57.98 ±0.4	64.22 ±0.35
DCPN	ResNet50 and PVT	**70.86 ±0.17**	**82.57 ±0.58**	**85.2 ±0.29**	**63.95 ±0.61**	**76.1 ±0.46**	**77.52 ±0.38**	**54.86 ±0.45**	**60.1 ±0.63**	**66.14 ±0.42**

The best and second-best results are highlighted.

TABLE II
COMPARISON AMONG THE CLASSIFICATION PERFORMANCES ACHIEVED WITH DIFFERENT BACKBONES ON THE CRCTP DATASET

	Backbone	Accuracy	AUC
CNN	VGG-16[61]	82.59±0.55	95.41±0.92
	VGG-19[61]	80.74±0.91	94.22±0.83
	DenseNet121[62]	85.26±0.29	98.56±0.79
	EfficientNet-B0[63]	84.28±0.38	97.11±0.43
	ResNet34[64]	84.96±0.47	97.42±0.48
	ResNet50[64]	**85.87±0.41**	**98.63±0.53**
Transformer	ViT_base[25]	84.74±0.49	98.52±0.67
	LeVit[65]	82.91±0.61	93.18±0.57
	BoTNet-50[66]	84.87±0.62	97.96±0.68
	Swin-T[67]	84.71±0.55	95.82±0.57
	PVT-small[48]	**85.15±0.49**	**98.60±0.41**

Bold values indicate the highest accuracy rate.

it more suitable for practical applications with limited computational resources. Overall, the dual-channel network-based feature mining process of the DCPN effectively improved the prototype representations and fully demonstrated the importance of multi-scale features in pathology image classification tasks.

D. Comparing the Classification Performances of Different Backbones

The experiment aimed to compare the classification performances achieved by different backbones on a pathology image dataset to determine the optimal feature extraction module, thereby supporting the design of a dual-channel feature embedding network. The CRCTP dataset (base dataset) was used for training and evaluating the models. In this experiment, we assessed 11 different backbones, including 6 classic CNN models and 5 transformer models. The overall experimental results are summarized in Table II. The results indicate that among the CNN approaches, ResNet50 achieved the best performance, with an accuracy and an AUC of 85.87% and 98.63%, respectively. Among the transformer methods, PVT-small achieved the best performance, with an accuracy and an AUC of 85.15% and 98.60%, respectively. Overall, the performances of the transformer models were still slightly inferior to that of ResNet50. We speculate that this may be related to the size of the CRCTP dataset, as transformer models typically adapt better to large-scale datasets and may not easily demonstrate their potential advantages on smaller datasets. Considering the above experimental results, we selected ResNet50 and PVT-small as the backbones for the dual-channel embedding network.

E. Impact of Pretraining on the Feature Encoding Performance of the PVT

In this experiment, we evaluated the impact of pretraining on the performance of the PVT model in few-shot classification tasks, and the results are shown in Table III. I Initially, we used the PVT solely as the encoder part of the prototypical network. The results indicate that pretraining the PVT in a self-supervised manner led to performance improvements across all the scenarios, with an average increase of 28.31%. This significant performance improvement is primarily attributed to self-supervised learning, which enabled the model to learn rich feature representations without the need for labeled data. This provided the PVT with a saturated feature space to alleviate its data-hungry nature, thus reducing the risk of overfitting in training tasks with limited sample data.

Subsequently, we constructed the DCPN by combining the PVT with ResNet and further explored the effect of pretraining within this architecture. Self-supervised pretraining also resulted in a performance increase for the DCPN, with an average improvement of 3.85%. More crucially, in the absence

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TABLE III
IMPACT OF SELF-SUPERVISED PRETRAINING ON PVT MODELS IN FEW-SHOT CLASSIFICATION TASKS

	Pretrained	Same domain			Near domain			Mixed domain		
		1-shot	5-shot	10-shot	1-shot	5-shot	10-shot	1-shot	5-shot	10-shot
ProtoNet	N	36.89±0.35	42.82±0.62	45.67±0.38	31.78±0.51	41.67±0.38	42.67±0.46	30.21±0.61	34.89±0.25	39.82±0.73
ProtoNet	Y	63.45±0.23	77.20±0.41	78.51±0.35	60.24±0.27	71.59±0.21	75.18±0.52	53.67±0.51	59.21±0.68	62.18±0.39
DCPN	N	62.52±0.39	78.02±0.57	78.82±0.46	61.21±0.56	70.92±0.49	75.83±0.69	53.95±0.44	59.11±0.58	62.28±0.49
DCPN	Y	70.86±0.17	82.57±0.58	85.2±0.29	63.95±0.61	76.1±0.46	77.52±0.38	54.86±0.45	60.1±0.63	66.14±0.42

TABLE IV
IMPACT OF MULTI-SCALE FEATURES ON THE PERFORMANCE ACHIEVED IN FEW-SHOT CLASSIFICATION TASKS

Feature	Same domain			Near domain			Mixed domain		
	1-shot	5-shot	10-shot	1-shot	5-shot	10-shot	1-shot	5-shot	10-shot
Global	63.42±0.23	73.89±0.61	76.21±0.32	60.58±0.61	71.01±0.49	75.14±0.52	53.22±0.67	59.03±0.43	61.59±0.72
Local	65.47±0.61	76.92±0.52	78.50±0.59	61.97±0.44	73.31±0.39	74.92±0.71	54.08±0.66	59.78±0.51	61.72±0.48
Mix	66.89±0.12	80.21±0.53	81.48±0.47	62.39±0.59	74.51±0.38	76.20±0.52	54.41±0.61	59.93±0.68	64.28±0.53
Multi-scale	70.86±0.17	82.57±0.58	85.2±0.29	63.95±0.61	76.1±0.46	77.52±0.38	54.86±0.45	60.1±0.63	66.14±0.42

of pretraining, the DCPN exhibited a significant advantage over the prototype network using the PVT alone, with a performance enhancement of 28.47%. We believe this gain stems from the inherent inductive bias of CNNs: local spatial continuity. When addressing high-resolution medical images, their local features may contain critical information about diseases or other medical conditions. Thus, CNNs provide the model with more robust and fine-grained representations of pathology images. These representations can compensate for the shortcomings of the PVT, effectively mitigating its high data demands.

F. The Necessity of Multi-scale Features

This experiment aimed to compare the impacts of features with different scales on the few-shot pathology image classification task, with the results summarized in Table IV. Here, "global" and "local" refer to the features extracted by the PVT and ResNet, respectively, while "mix" represents the concatenated global and local features obtained after performing dimensionality reduction through PCA. Furthermore, "multi-scale" denotes the use of all three types of features in concert for the few-shot pathology image classification task, as proposed in our DCPN architecture.

In the same-domain scenario, the local features achieved accuracies of 65.47%, 76.92%, and 78.5% in the 1-shot, 5-shot, and 10-shot settings, respectively, clearly outperforming the global features, which yielded 63.42%, 73.89%, and 76.21% accuracies, respectively. Further observations in the near-domain and mixed-domain scenarios also showed the superior performance of the local features to that of the global features. This indicates that local features have better discriminability in few-shot pathology image classification tasks. The mixed features performed better than either the local or global features alone in all scenarios, suggesting that a combination of features with different scales can provide more comprehensive image representations. The feature visualization in Fig. 6 also clearly shows the differences between the PVT and ResNet features

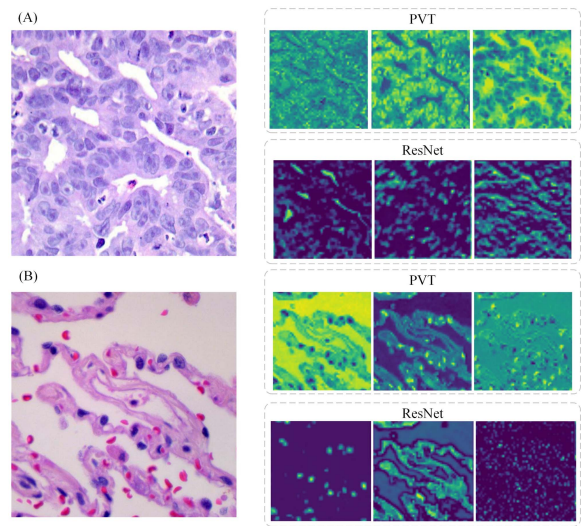

Fig. 6. Visualization of pathological features encoded by the PVT and ResNet. (a) and (b) represent two randomly selected pathological images and their corresponding feature visualization maps.

and that their features are complementary. The performance of the multi-scale features was outstanding, and we believe that there are several reasons for this finding. 1) PCA-based dimensionality reduction may cause the concatenated mixed features to lose some discriminative information that is helpful for classification purposes. 2) Multi-scale features dynamically adjust the influence of the features at each scale on the resulting classification decision according to the confidence level of the classifier through a soft voting mechanism, achieving optimal classification results.

G. The Impacts of Different Distance Evaluation Methods on the DCPN

This experiment aimed to compare whether the cosine similarity or Euclidean distance is more suitable for assessing the

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TABLE V
IMPACTS OF DIFFERENT DISTANCE METRICS METHODS ON THE DCPN MODEL

Method	Same domain			Near domain			Mixed domain		
	1-shot	5-shot	10-shot	1-shot	5-shot	10-shot	1-shot	5-shot	10-shot
Cosine similarity	64.37±0.43	75.49±0.39	80.67±0.44	57.91±0.51	72.21±0.39	74.15±0.52	51.32±0.61	57.80±0.85	62.69±0.69
Euclidean distance	70.86±0.17	82.57±0.58	85.2±0.29	63.95±0.61	76.1±0.46	77.52±0.38	54.86±0.45	60.1±0.63	66.14±0.42

TABLE VI
DATA PARTITIONING DETAILS OF THE BLOOD DATASET

Dataset	Meta-training	Meta-validation	Meta-testing
BLOOD	Eosinophil (600) Erythroblast (600) Neutrophil (600) Platelet (600)	IG-ig (151) IG-mmy (600) IG-my (600) IG-pmy (592)	Basophil (600) Lymphocyte (600) Monocyte (600)

distances between the query samples and prototype features produced within prototype-based few-shot classification methods for pathology images. The results in Table V indicate that across all three scenarios, employing the Euclidean distance as the metric yielded superior performance to that of cosine similarity. Particularly in the "same-domain" scenario, the 5-shot classification task implemented using Euclidean distance achieved an accuracy of 82.57%, which was significantly higher than the 75.49% accuracy achieved using cosine similarity. Moreover, in the 10-shot task conducted in the "mixed-domain" scenario, using the Euclidean distance achieved an accuracy of 66.14%, showing a clear advantage over the 62.69% accuracy attained with cosine similarity. This conclusion is consistent with research findings in the field of natural images [14].

Cosine similarity measures the similarity between two feature vectors by calculating the cosine of the angle between them, primarily reflecting direction differences rather than scale differences. In contrast, the Euclidean distance involves not only the directional differences among vectors but also their scale differences, representing the absolute numerical discrepancy between feature vectors. This suggests that the Euclidean distance metric may provide more information when evaluating the differences between samples and prototype representations. It is noteworthy that when the Bregman divergence is used as the metric, the prototype network can be seen as a probabilistic distribution estimation process over the support set. However, the cosine distance is not a Bregman divergence and hence does not possess its corresponding properties. This may explain why, in this study, the performance of the squared Euclidean distance (a type of Bregman divergence) exceeded that of the cosine distance.

H. Generalization Performance Achieved on Non-Pathology Image Datasets

To further validate the efficacy of the proposed DCPN method in the domain of nonpathological medical imaging, we incorporated a dataset consisting of microscopic peripheral blood cell images as the subject of our study. The specific details of the data partitioning strategy described in reference [16] are presented in Table VI. Moreover, this study also compared 13 typical FSL algorithms, with detailed experimental results listed in Table VII. Notably, the results of MAML, ProtoNet, RelationNet [68], Versa [69], R2D2 [70], ANIL [71], MTL [72], and GDSL [16] were obtained directly from the literature [16]. he experimental results demonstrate that the DCPN algorithm achieved superior performance on the microscopic peripheral blood cell image dataset. In the 1-shot, 5-shot, and 10-shot tasks, the accuracy of the proposed method increased by an average of 1.17% ($p<0.05$) over that of the second-best method, which is statistically significant. This finding confirms that the DCPN algorithm not only has a competitive advantage in the field of pathology image analysis but also exhibits significant strengths on other nonpathological medical imaging datasets.

TABLE VII
PERFORMANCE COMPARISON BETWEEN THE DCPN AND OTHER METHODS ON THE BLOOD DATASET

Method	blood		
	1-shot	5-shot	10-shot
MAML[8]	54.37±0.89	56.54±0.75	57.69±0.66
ProtoNet[14]	57.4±1.16	71.62±0.95	73.23±0.55
RelationNet [68]	56.62±0.99	61.14±0.48	66.79±0.66
Versa [69]	59.63±0.79	62.79±0.53	64.63±0.49
R2D2[70]	55.93±0.79	68.06±0.60	**82.66±0.47**
ANIL[71]	61.28±0.81	72.62±0.61	74.36±0.64
MTL[72]	59.07±1.04	69.36±0.65	70.55±0.55
CGRN[59]	61.59±0.67	68.67±0.74	79.58±0.69
LDP-Net[60]	62.97±0.74	**73.31±0.59**	81.45±0.67
Latent Augmentation[58]	61.58±0.71	70.22±0.71	75.43±0.78
Histology Siamese Network[19]	60.42±0.83	68.56±0.59	72.29±0.91
Multi-ProtoNets[21]	63.1±0.75	72.44±0.62	80.01±0.76
GDSL[16]	**63.47±0.94**	71.97±0.66	76.21±0.50
DCPN	**64.52±0.72**	**74.67±0.64**	**83.76±0.59**

The best and second-best results are highlighted.

I. Parameter Sensitivity Analysis

To further explore the specific impacts of different hyperparameter settings on the performance of the DCPN method, this study initially analyzed the effects of different learning rates, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}, on the accuracy achieved by the model on the CRCTP dataset. As depicted in Fig. 7(a), the model accuracy peaked when the learning rate was set to 10^{-4}. Subsequently, we compared three different optimizers: SGD, RMSprop, and Adaw. According to the experimental data in Fig. 7(b), the Adaw optimizer performed best in terms of model accuracy. Finally, by plotting the accuracy variation curve

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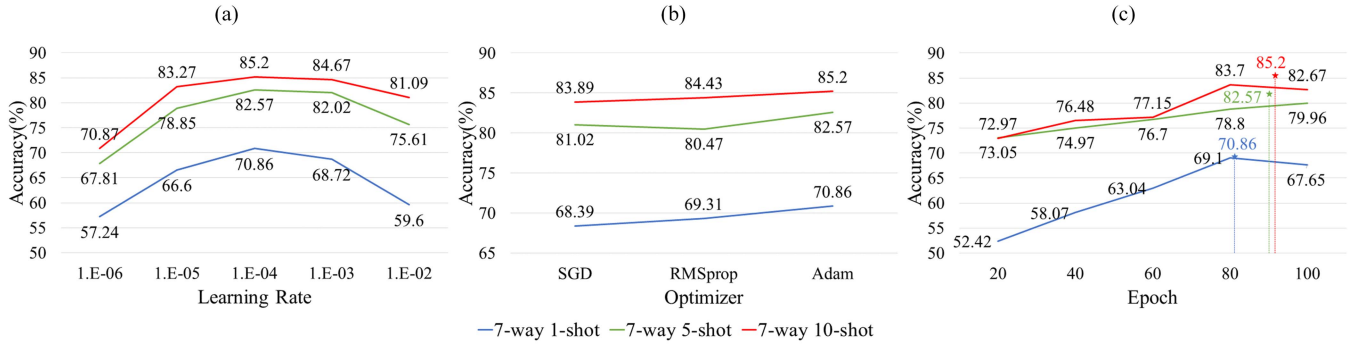

Fig. 7. Impacts of different hyperparameters on the DCPN.

exhibited by the DCPN method across different training epochs, it was observed that for the 1-shot, 5-shot, and 10-shot task settings, the model achieved its optimal performance at the 81st, 89th, and 91st training epochs, respectively. These results are clearly indicated by the blue, green, and red stars in Fig. 7(c).

VI. Conclusion

In this study, we introduced a novel dual-channel prototype network (DCPN) that was aimed at addressing the problem of classifying pathology images in few-shot learning scenarios. Given the inherent long-tailed distribution characteristics of pathology images and the scarcity of annotated samples, our designed DCPN model integrates a pyramid vision transformer (PVT) and a convolutional neural network (CNN), effectively mining multi-scale and fine-grained pathological features. This strategy significantly enhances the generalizability of the constructed prototype representations. The experimental results obtained on three public pathological datasets simulating real-world clinical problems confirmed the significant advantage provided by the DCPN in few-shot pathology image classification tasks. Particularly, in same-domain tasks, its performance was even comparable to that of traditional supervised learning methods. However, despite achieving SOTA accuracy, the DCPN model exhibited a noticeable decrease in performance as the difficulty of the cross-domain tasks increased, highlighting the generalization challenges faced due to data heterogeneity. Future work will focus on exploring large-scale self-supervised pretraining and domain adaptation strategies to construct more efficient algorithms for few-shot pathology image classification tasks, aiming to enhance the robustness and adaptability of the model across diverse clinical settings.

Conflict and Interest

The authors declare that they don't have conflict and interest.

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